Section 1 — Probability and Statistics

Lecture 1 — Set Theory Fundamentals

Beginner-friendly, exam-oriented study notes · no prior set theory assumed

LEARNING OBJECTIVES

By the end of this lecture you should be able to:

- define a **set**, decide whether a collection is well-defined, and use the symbols $\in$ and $\notin$ correctly;
- write a set in **roster**, **set-builder** and **descriptive** form, and convert between them;
- recognise **empty, singleton, finite, infinite, equal, equivalent, disjoint** sets, subsets, proper subsets, and **intervals of real numbers**;
- use $\subseteq$, $\subset$, $=$, $\neq$ and the cardinality $|S|$ accurately, including the fact that $\varnothing \subseteq S$ for every set S;
- explain what a **universal set** U is and why it depends on the problem;
- find the **complement** $A^c = U \setminus A$ of a set and draw it;
- avoid the standard exam traps in all of the above.

How to read the boxes. Each box type has its own border style, so the meaning is clear even in a black-and-white photocopy:

Definition (solid, thick)	**Notation** (double line)	**Worked example** (thin line)	**Trap / mistake** (dashed line)	**Exam tip** (dotted line)

1. Introduction to Sets

A **set** is the most basic building block of mathematics. In probability, outcomes, events and sample spaces are all described using sets, so we start here and build up carefully.

DEFINITION Set

A **set** is a *well-defined collection of distinct objects*. The objects are called the **elements** (or **members**) of the set.

- **Well-defined** — for any object, we can decide without argument whether it is in the set or not.
- **Distinct** — each element counts once. Listing an element twice changes nothing: {1, 1, 2} = {1, 2}.
- **Order does not matter** — {1, 2, 3} and {3, 1, 2} are the same set.

1.1 Examples from computing

Collection	Set?	Reason
The four inbox categories: Primary, Social, Promotions, Spam	**Yes**	Four named objects; membership is clear.
Records of a dataset whose age is greater than 30	**Yes**	Every record can be tested against the condition.
Class labels of a digit classifier: 0, 1, 2, …, 9	**Yes**	Ten clearly listed machine-learning classes.
Even integers between 1 and 10 inclusive	**Yes**	The set is {2, 4, 6, 8, 10}.
“Good emails”	**No**	“Good” is an opinion; there is no objective test.
“Large datasets”	**No**	How large is large? The boundary is not fixed.

A vague collection becomes a set once we fix a precise rule, for example “datasets with more than 1 000 000 rows”.

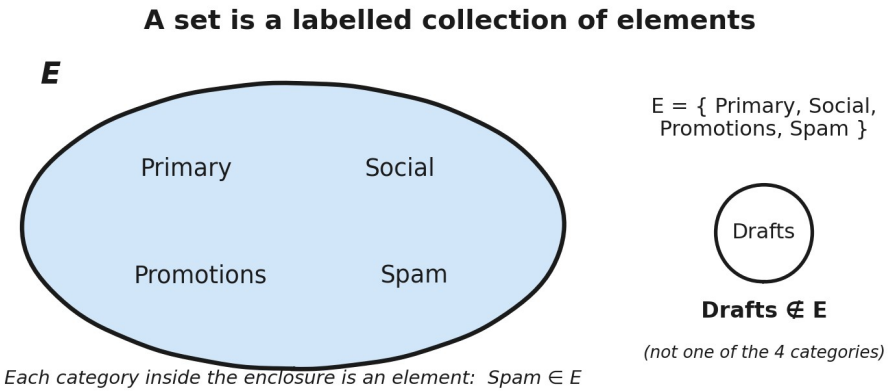


Figure 1.1 — A set drawn as an enclosure. E holds the four email categories; "Drafts" is outside, so Drafts $\notin$ E.

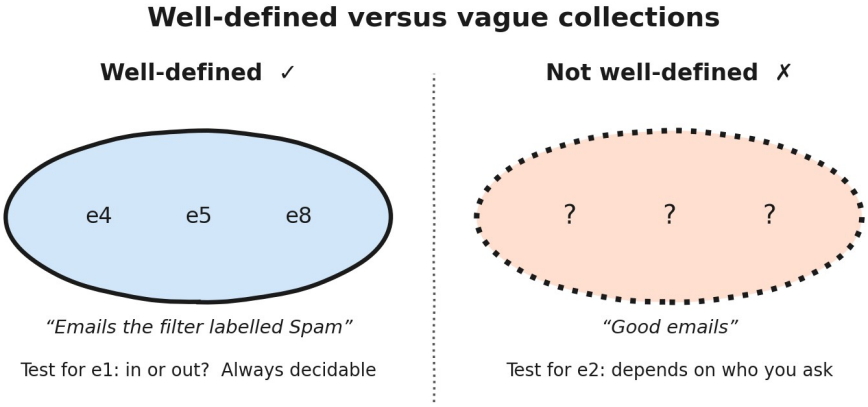


Figure 1.2 — A well-defined collection has a clear in-or-out test (solid outline). A vague one does not (dotted outline, "?").

WORKED EXAMPLE **Example 1.1 — Is it a set?**

Decide whether "the emails in a folder that the spam filter has labelled Spam" is a set.

Step 1. Pick any email and ask: *has the filter labelled it Spam?*

Step 2. The answer is always a definite yes or no — no opinion is needed.

Conclusion: it is a well-defined set. By contrast, "the most annoying emails" is **not** a set.

COMMON MISTAKE / EXAM TRAP **Lists are not sets**

A dataset column [3, 5, 3, 7] is a *list* with a repeated value. The **set** of its values is {3, 5, 7}, which has 3 elements. Sets ignore repeats and order.

2. Set Notation

Sets are named with **capital letters** (A, S, T, E, …). Elements are written inside **curly braces** { } separated by commas. Individual elements are usually lower-case letters or plain values.

NOTATION **Membership symbols**

x ∈ S — read "x belongs to S" or "x is an element of S".

x ∉ S — read "x does not belong to S". The slash through ∈ means "not".

Symbol	Example, with S = {2, 4, 6, 8}	Read as	True / False
{ }	S = {2, 4, 6, 8}	"S is the set whose elements are 2, 4, 6, 8"	—
∈	4 ∈ S	4 is an element of S	True
∈	5 ∈ S	5 is an element of S	False — 5 is not listed
∉	5 ∉ S	5 is not an element of S	True
∉	8 ∉ S	8 is not an element of S	False — 8 is listed

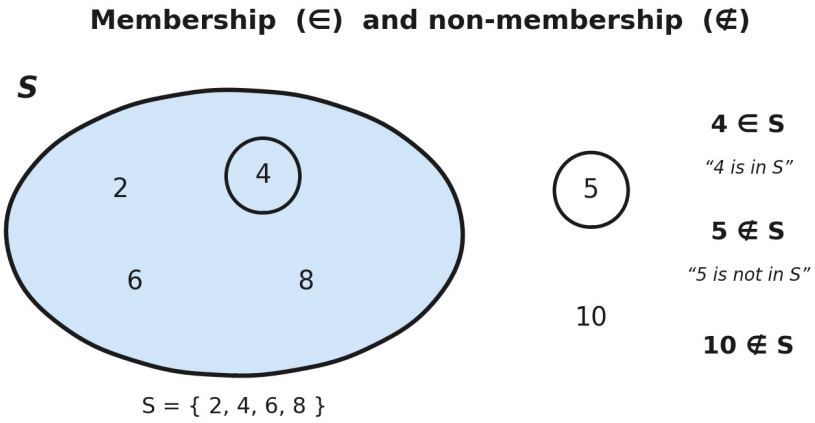


Figure 2.1 — Membership: 4 is inside the enclosure (4 ∈ S); 5 and 10 are outside (5 ∉ S, 10 ∉ S).

2.1 An element is not the same as a set

The number 4 is an element. The expression {4} is a *set* whose only element is 4. They are different objects. For S = {2, 4, 6, 8}:

- **4 ∈ S** is true (4 is an element of S).
- **{4} ∈ S** is false (the elements of S are numbers, not sets).
- **{4} ⊆ S** is true (the set {4} is a *subset* of S; see Topic 5).

Rule of thumb: ∈ links an *element* to a *set*. ⊆ and ⊂ link a *set* to a *set*.

WORKED EXAMPLE **Example 2.1 — Image-classifier classes**

Let C = {cat, dog, bird} be the class labels of an image classifier.

- dog ∈ C — **True** (dog is listed).
- fish ∉ C — **True** (fish is not listed).
- {dog} ∈ C — **False** ({dog} is a set; the elements of C are labels).
- |C| = 3 — C has three elements.

COMMON MISTAKE / EXAM TRAP **Mixing up ∈ and ⊆**

Writing "4 ⊆ S" or "{4} ∈ S" is wrong. Ask yourself: *is the left side a single element or a set?* Single element → ∈. Set → ⊆ or ⊂.

3. Ways of Representing a Set

The same set can be written in three ways. Choose the one that is clearest for the problem.

Form	Idea	Example (the same set B)
Roster (tabular)	List every element inside braces	B = {2, 4, 6, 8}
Set-builder	Give a rule that elements must satisfy	B = {x ∈ ℤ : x is even and 2 ≤ x ≤ 8}
Descriptive (statement)	Describe the set in words	B = the set of even integers from 2 to 8, inclusive

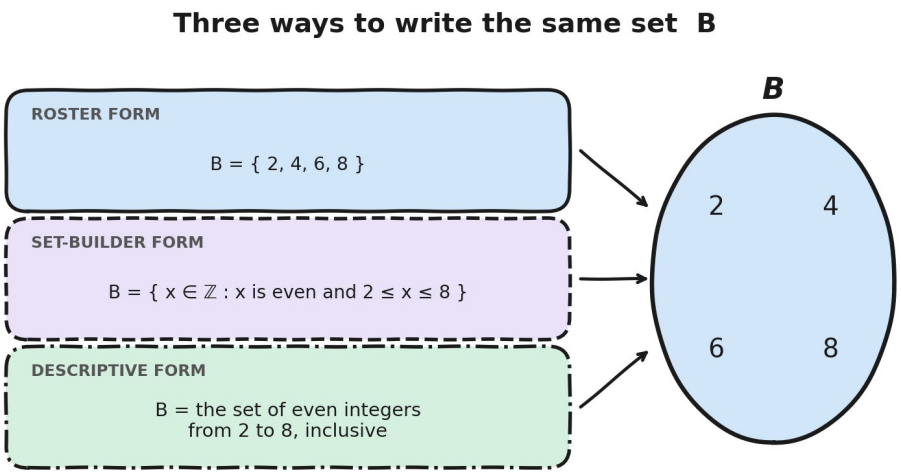


Figure 3.1 — Roster, set-builder and descriptive forms are three descriptions of one and the same set B.

3.1 Reading set-builder notation

Take B = { x ∈ ℤ : x is even and 2 ≤ x ≤ 8 }. Each part has a job:

Part	Meaning
{ }	Braces: the set is *everything described inside*.
x	The **variable**: a placeholder for a typical element.
∈ ℤ	x is chosen from the integers (the **domain**).
: (or |)	The **separator**, read "such that".
x is even and 2 ≤ x ≤ 8	The **condition**: x must satisfy *every* part of it.
"and" or a comma	Both conditions must hold at the same time.

NOTATION **Number sets used in these notes**

ℕ = {1, 2, 3, …} the natural numbers (*convention:* some textbooks include 0; here ℕ starts at 1).

ℤ = {…, −2, −1, 0, 1, 2, …} the integers. ℝ = all real numbers (every point of the number line).

Another common style writes the rule as an expression: {n^2 : n ∈ ℕ, n ≤ 4} means "the values n^2 as n runs over 1, 2, 3, 4", i.e. {1, 4, 9, 16}. Some books use | instead of : — they mean the same.

3.2 Converting between the forms

WORKED EXAMPLE **Example 3.1 — Roster ⇄ set-builder ⇄ descriptive**

(a) Roster → set-builder. Write {1, 4, 9, 16, 25} in set-builder form.

Step 1. Spot the pattern: 1 = 1^2, 4 = 2^2, 9 = 3^2, 16 = 4^2, 25 = 5^2.

Step 2. Write the rule: {n^2 : n ∈ ℕ and n ≤ 5}. Check: n = 1,…,5 gives exactly the five elements. ✓

(b) Set-builder → roster. List {x ∈ ℤ : −2 ≤ x < 3}.

Step 1. Integers start at −2 (included) and go up to, but **not including**, 3 (because < is strict).

Step 2. Roster: {−2, −1, 0, 1, 2}, which has 5 elements.

(c) Descriptive → set-builder → roster. A dataset D = {r1, …, r6} has ages 25, 34, 41, 30, 29, 52. Describe "records older than 30".

Step 1. Set-builder: O = {r ∈ D : age(r) > 30}.

Step 2. Test each record: r1 (25) no, r2 (34) yes, r3 (41) yes, r4 (30) **no** — not strictly greater, r5 (29) no, r6 (52) yes.

Roster: O = {r2, r3, r6}.

COMMON MISTAKE / EXAM TRAP **Typical slips**

- Round brackets (1, 2, 3) mean an **ordered list**, not a set; sets use braces { }.
- Forgetting the domain: "{x : x is even}" is unclear unless we say *even what* ($x \in \mathbb{Z}$? $x \in \mathbb{N}$?).
- Mixing strict (<, >) and non-strict (≤, ≥) inequalities — the boundary element is in or out.
- Using "…" when the pattern is not obvious. Use it only when the pattern is clear, e.g. {1, 2, 3, …}.

4. Types of Sets

4.1 Empty (null) set

DEFINITION **Empty set**

The **empty set** has no elements. Notation: **∅** or **{ }**. Its cardinality is $|\emptyset| = 0$.

Examples: $\{x \in \mathbb{R} : x^2 = -1\} = \emptyset$ (no real number squares to −1); $\{x \in \mathbb{Z} : 2 < x < 3\} = \emptyset$ (no integer lies strictly between 2 and 3). In the email model of §4.8, "emails labelled both Spam and Primary" is empty, because each email gets exactly one label.

4.2 Singleton set

DEFINITION **Singleton set**

A **singleton** has exactly **one** element, e.g. {5}. Cardinality $|\{5\}| = 1$.

Examples: $\{x \in \mathbb{Z} : 2x = 8\} = \{4\}$; Spam = {e5} in the email batch of §4.8.

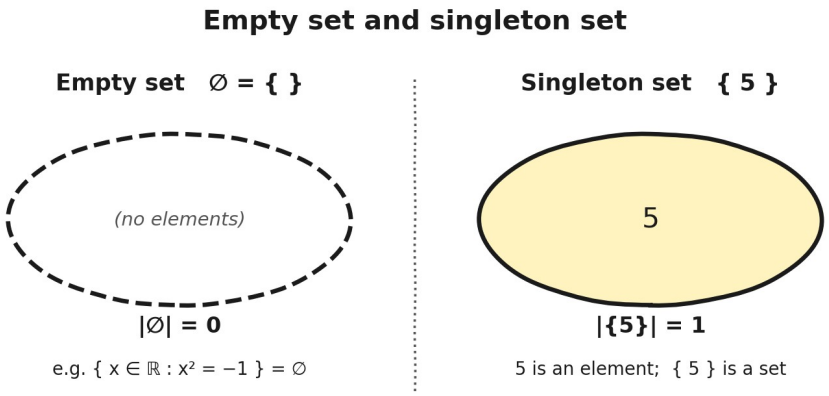


Figure 4.1 — Empty set (dashed, nothing inside) and singleton set (one element). Note that 5 and {5} are different objects.

COMMON MISTAKE / EXAM TRAP **∅, {0} and {∅} are all different**

- ∅ has **0** elements. {0} has **1** element (the number 0). So $\{0\} \neq \emptyset$.
- {∅} is a set containing one element (the empty set), so $|\{\emptyset\}| = 1$. It is **not** empty. Write ∅ — not {∅} — for "no elements".

4.3 Finite and infinite sets

DEFINITION **Finite and infinite**

A set is **finite** if its elements can be counted and the counting ends: $|S|$ is a whole number 0, 1, 2, … . It is **infinite** if the counting never ends.

∅ and every singleton are finite. The set of class labels {0, …, 9} is finite with 10 elements. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z}$, and the even integers are infinite. A *very large* set is still finite: the set of all 64-bit unsigned integers has 2^{64} elements, a huge but whole number.

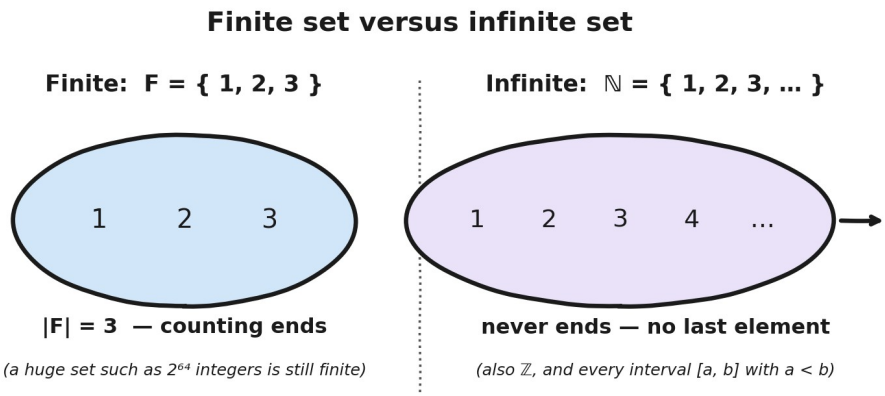


Figure 4.2 — F = {1, 2, 3} is finite (|F| = 3). ℕ is infinite: the arrow shows the listing never ends.

4.4 Equal sets

DEFINITION **Equal sets**

Sets A and B are **equal**, written **A = B**, if they have exactly the same elements: every element of A is in B **and** every element of B is in A.

Order and repetition are irrelevant: {1, 2, 3} = {3, 1, 2} = {1, 1, 2, 3}. Also $\{x \in \mathbb{Z} : x^2 = 4\} = \{-2, 2\}$. If even one element differs, the sets are not equal: {1, 2} ≠ {1, 2, 3}. (See the right-hand panel of Figure 4.4.)

4.5 Equivalent sets

DEFINITION **Equivalent sets**

Sets A and B are **equivalent**, written **A ~ B**, if they have the same number of elements: |A| = |B|. Equivalent sets can be paired off one-to-one.

Example: A = {a, b, c} and B = {1, 2, 3} are equivalent (both have 3 elements) but **not equal**. *Equal* ⇒ *equivalent*, but *equivalent* ⇏ *equal*. (Textbooks vary: some write A ≈ B or |A| = |B| instead of A ~ B. For infinite sets, "equivalent" means a one-to-one pairing exists; we only use the finite case here.)

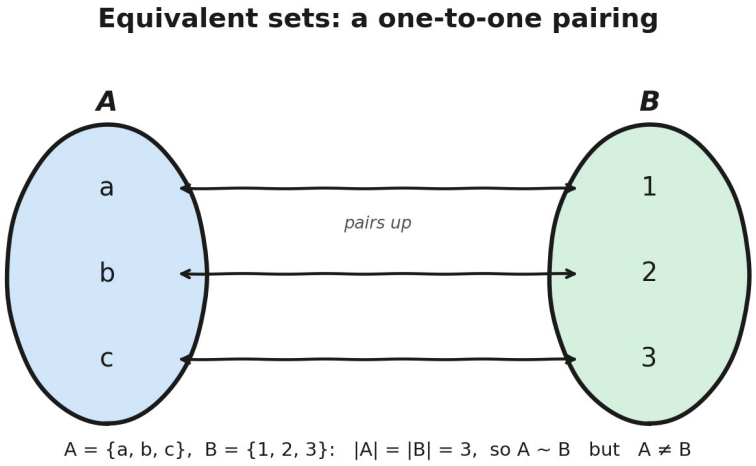


Figure 4.3 — Equivalent sets: each element of A is paired with exactly one element of B, so |A| = |B| = 3, yet A ≠ B.

4.6 Subsets and proper subsets

DEFINITION **Subset and proper subset**

A ⊆ B ("A is a subset of B") means every element of A is also an element of B. A may equal B.

A ⊂ B ("A is a proper subset of B") means A ⊆ B **and** A ≠ B; so B has at least one element that A lacks.

NOTATION **Convention used in these notes**

Here **$\subset$ always means proper subset** and **$\subseteq$ allows equality**. Some textbooks use $\subset$ for "subset, possibly equal" and write $\subsetneq$ for "proper subset". Always check which convention your textbook or exam uses.

Example: with A = {1, 2} and B = {1, 2, 3, 4}: $A \subseteq B$ and $A \subset B$ are both true. But $B \subseteq B$ is true while $B \subset B$ is false. Useful chain: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R}$. The empty set is a subset of every set (proved in §5.3). *Good to know:* a set with n elements has $\mathbf{2^n}$ subsets; for {a, b} they are ∅, {a}, {b}, {a, b}.

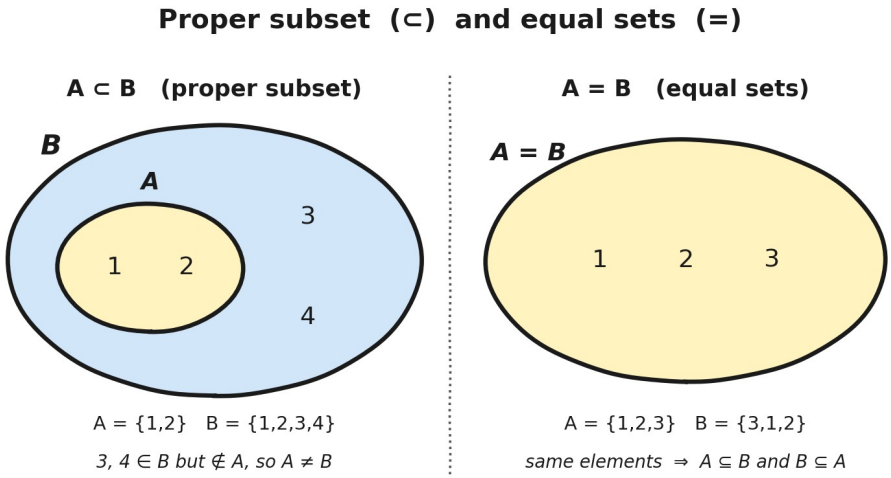


Figure 4.4 — Left: $A \subset B$ (A is inside B and B has extra elements 3, 4). Right: A = B (same elements, so $A \subseteq B$ but not $A \subset B$).

4.7 Disjoint sets

DEFINITION **Disjoint sets**

Two sets are **disjoint** if they have **no element in common**. A family of sets is *pairwise disjoint* if every two of them are disjoint.

Examples: {1, 2} and {3, 4} are disjoint. {1, 2} and {2, 3} are **not** (2 belongs to both). ∅ is disjoint from every set.

WORKED EXAMPLE **Example 4.1 — Email categories (the running example)**

A filter processes six emails e1,…,e6 and assigns **exactly one** label to each: Primary = {e1, e4}, Social = {e2}, Promotions = {e3, e6}, Spam = {e5}.

Modelling assumption: every email receives exactly one label, so no email appears in two categories and the four sets are pairwise disjoint. (If emails could carry several labels, the sets could overlap and would *not* be disjoint.)

Where do probabilities come in? A real classifier produces a score per category and a *decision rule* (for instance "take the highest score") picks the single label. The sets are the groups of emails *after* that rule is applied; a probability alone does not put an email in a set.

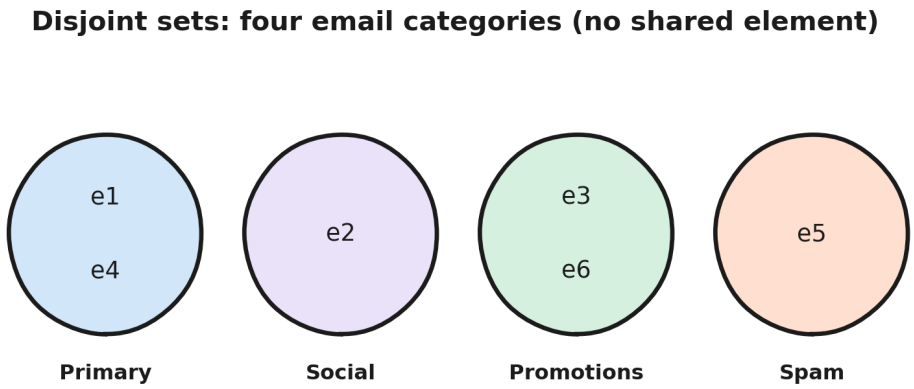


Figure 4.5 — Four separate enclosures: no element is shared, so the categories are pairwise disjoint (under the one-label assumption).

4.8 Continuous intervals: subsets of the real numbers

The real numbers $\mathbb{R}$ fill the whole number line with no gaps. A **real interval** is the set of all real numbers between two endpoints.

Interval	Set-builder form	Endpoints
[a, b]	$\{x \in \mathbb{R} : a \le x \le b\}$	both included — **closed**
(a, b)	$\{x \in \mathbb{R} : a < x < b\}$	both excluded — **open**
[a, b)	$\{x \in \mathbb{R} : a \le x < b\}$	a included, b excluded — half-open
(a, b]	$\{x \in \mathbb{R} : a < x \le b\}$	a excluded, b included — half-open
[a, ∞)	$\{x \in \mathbb{R} : x \ge a\}$	a included; unbounded to the right
(−∞, b)	$\{x \in \mathbb{R} : x < b\}$	b excluded; unbounded to the left

NOTATION **Reading brackets and dots**

Square bracket [ ] = endpoint **included** = **filled dot** ●. Round bracket ( ) = endpoint **excluded** = **hollow dot** ○. The symbol ∞ is not a number, so it always takes a round bracket.

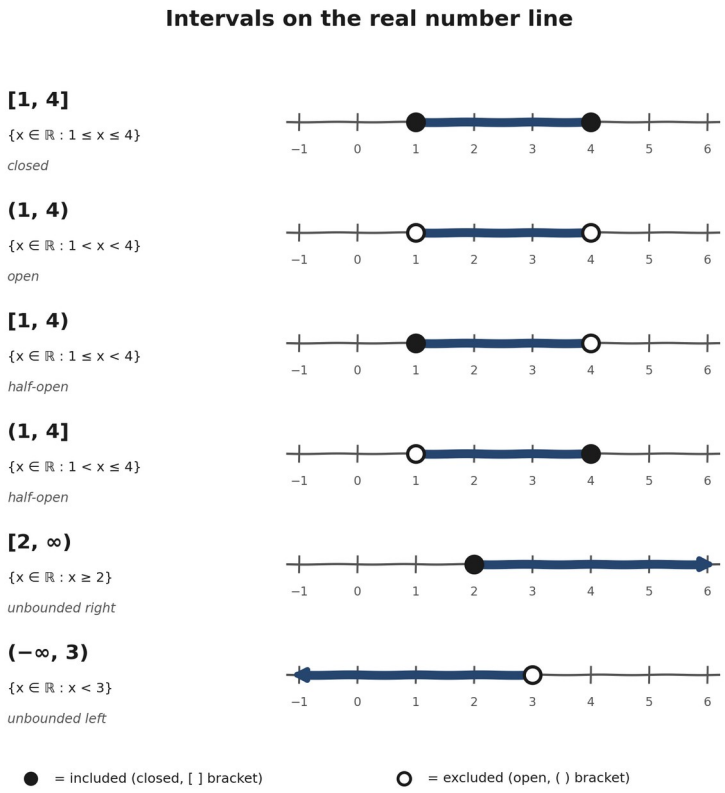


Figure 4.6 — Intervals on the real number line. Filled dots are included endpoints; hollow dots are excluded endpoints. Arrows mean the interval continues forever.

KEY IDEA **Continuous intervals are not a third category next to finite/infinite**

- "Continuous" describes a *real interval*. Every set is finite **or** infinite; an interval [a, b] with a < b is one particular kind of **infinite** set (between any two points there is another).
- **Not every infinite set is continuous.** $\mathbb{N} = \{1, 2, 3, \ldots\}$ is infinite but has gaps (nothing between 2 and 3).
- [1, 4] contains 2.5 and 3.99; $\{x \in \mathbb{Z} : 1 \le x \le 4\} = \{1, 2, 3, 4\}$ contains only whole numbers.
- Edge cases: [3, 3] = {3} is a singleton; (3, 3) = ∅.

WORKED EXAMPLE **Example 4.2 — Intervals in machine learning**

- A probability value p satisfies $0 \le p \le 1$, so $p \in$ **[0, 1]** (closed: 0 and 1 are possible).
- A sigmoid output is strictly between 0 and 1, so it lies in **(0, 1)** (open).
- A pixel intensity stored as an 8-bit integer lies in {0, 1, …, 255}: **finite and discrete**, *not* an interval.

COMMON MISTAKE / EXAM TRAP **Mistakes with intervals**

- Saying "every infinite set is continuous" — false ($\mathbb{N}$, $\mathbb{Z}$).
- Confusing [1, 4] (all reals from 1 to 4) with {1, 2, 3, 4} (four integers).
- Wrong endpoint: x < 4 gives (…, 4) with a **hollow** dot; x ≤ 4 gives (…, 4] with a **filled** dot.
- (1, 4) can also mean an ordered pair; the context tells you which.

5. Set Relationships and Notation

Notation	Read as	Meaning	Example
$S \subseteq T$	S is a subset of T	every element of S is in T (S may equal T)	$\{1, 2\} \subseteq \{1, 2, 3\}$ ✓
$S \subset T$	S is a proper subset of T	$S \subseteq T$ **and** $S \neq T$	$\{1, 2\} \subset \{1, 2, 3\}$ ✓
$S \not\subset T$	S is not a subset of T	some element of S is not in T	$\{1, 5\} \not\subset \{1, 2, 3\}$ ✓
$S = T$	S equals T	exactly the same elements	$\{1, 2, 3\} = \{3, 2, 1\}$ ✓
$S \neq T$	S is not equal to T	they differ in at least one element	$\{1, 2\} \neq \{1, 2, 3\}$ ✓
$\lvert S \rvert$ or n(S)	cardinality of S	number of distinct elements of a finite set	$\lvert\{a, b, c\}\rvert = 3$

5.1 Cardinality

DEFINITION Cardinality

The **cardinality** of a finite set S is the number of distinct elements in it. These notes write it **$|S|$**; many school textbooks write **n(S)** — the two mean the same thing.

Examples: $|\{a, b, c\}| = 3$; $|\varnothing| = 0$; $|\{\varnothing\}| = 1$; $|\{1, 1, 2\}| = 2$ (repeats are counted once). For an infinite set we simply say its cardinality is infinite.

5.2 Element of versus subset of

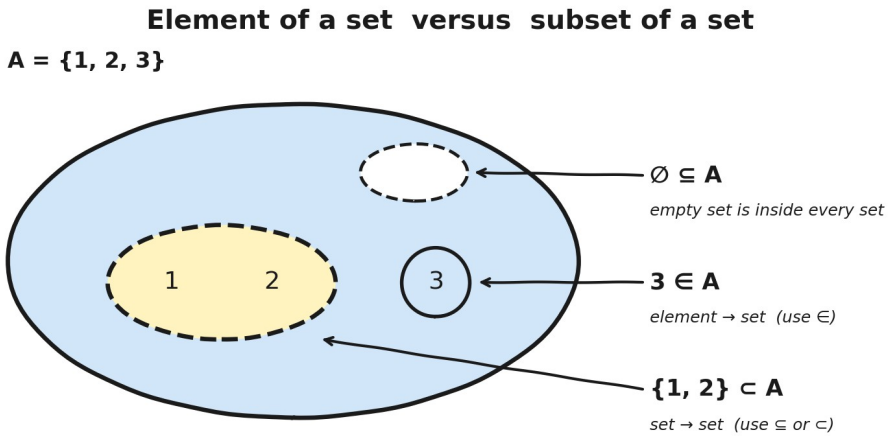


Figure 5.1 — $\in$ compares an element with a set ($3 \in A$). $\subseteq$ and $\subset$ compare two sets ($\{1, 2\} \subset A$). The empty set sits inside every set.

Statement (A = {1, 2, 3})	Truth	Why
$2 \in A$	True	2 is listed in A.
$\{2\} \in A$	False	The elements of A are numbers, not sets.
$\{2\} \subseteq A$	True	The only element of {2} is 2, which is in A.
$\{1, 2\} \subset A$	True	Subset, and A has the extra element 3.
$A \subseteq A$	True	Every element of A is in A.
$A \subset A$	False	A = A, so it is not a *proper* subset.
$\varnothing \subseteq A$	True	See §5.3.
$\varnothing \in A$	False	$\varnothing$ is not one of the listed elements 1, 2, 3.
$\{1, 5\} \subseteq A$	False	$5 \in \{1, 5\}$ but $5 \notin A$.
$4 \subseteq A$	Meaningless	4 is not a set, so $\subseteq$ cannot be used with it.

5.3 The empty set is a subset of every set

KEY IDEA **Why $\varnothing \subseteq S$ for every set S**

A statement $S_1 \subseteq T$ can fail **only** if we can find an element of S_1 that is not in T. The empty set has **no elements**, so there is nothing that could fail. Therefore $\varnothing \subseteq T$ is true for every set T. (It is a *proper* subset whenever T is not empty.)

WORKED EXAMPLE **Example 5.1 — Relationships among A, B, C**

Let A = {1, 2}, B = {1, 2, 3} and C = {3, 2, 1}.

Step 1. **B = C**: same elements, order is irrelevant. So $B \subseteq C$ and $C \subseteq B$, and $|B| = |C| = 3$.

Step 2. **$A \subset B$**: both 1 and 2 are in B, and $3 \in B$ but $3 \notin A$, so $A \neq B \rightarrow$ proper subset.

Step 3. **$A \neq B$**, and **$A \subset C$** too (C is the same set as B).

Step 4. **$B \not\subset A$**, because $3 \in B$ but $3 \notin A$.

Step 5. **$\varnothing \subseteq A, B, C$** — always true.

EXAM TIP **How to test a relationship**

- **$S \subseteq T$:** take each element of S and look for it in T. One miss $\Rightarrow$ not a subset.
- **S = T:** check *both* $S \subseteq T$ and $T \subseteq S$.
- **$S \subset T$:** check $S \subseteq T$, then find one element of T that is not in S.

COMMON MISTAKE / EXAM TRAP **Traps with $\subset$ and $\in$**

- "$A \subset A$" is **false** with our convention (a set is not a *proper* subset of itself).
- Confusing $|\{1, 2, 3\}| = 3$ with the set {3}; cardinality is a **number**, not a set.
- $\varnothing \subseteq A$ is true but $\varnothing \in A$ is usually false.

6. The Universal Set

DEFINITION **Universal set**

The **universal set**, written **U**, is the set of *all elements under consideration in a given problem*. Every set in that problem is a **subset of U**.

U is **not one fixed set** for all of mathematics. You choose it from the context of the problem. The same set A can sit inside different universal sets.

Situation	Universal set U	A subset A of U
Batch of six emails	U = {e1, e2, e3, e4, e5, e6}	A = Primary = {e1, e4}
Dataset of 10 records	U = {r1, r2, …, r10}	A = records labelled "positive"
Integers from 1 to 20	U = {1, 2, …, 20}	A = primes in U = {2, 3, 5, 7, 11, 13, 17, 19}
Probability value of a model	U = [0, 1]	A = [0.5, 1] (scores at or above 0.5)

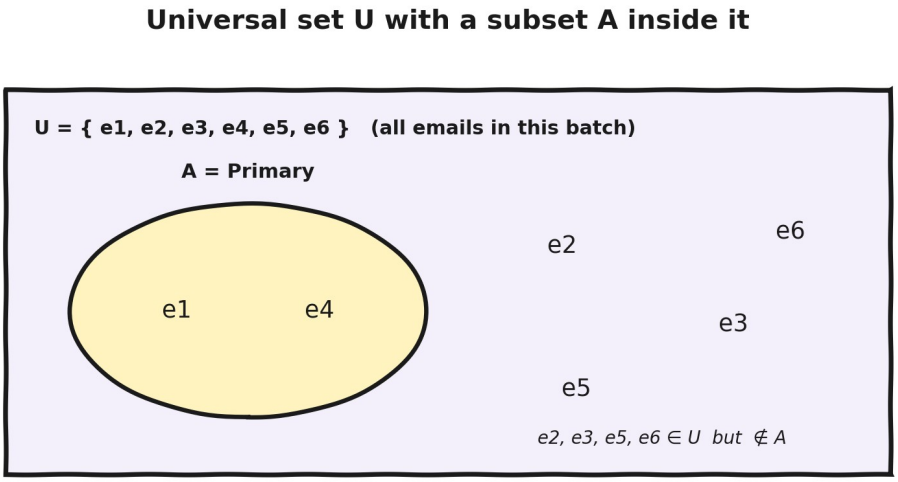


Figure 6.1 — U is the outer rectangle (all six emails). A = Primary is a subset drawn inside it. The other emails are in U but not in A.

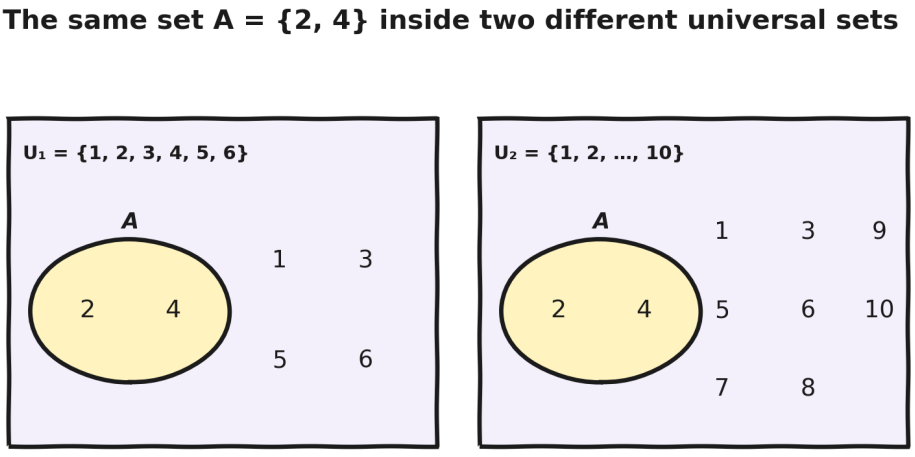


Figure 6.2 — The same A = {2, 4} lies in different universal sets. Choosing U decides which other elements are "in the picture".

WORKED EXAMPLE **Example 6.1 — Choosing U**

"Let A be the set of even numbers 2, 4, 6." What could U be?

- If we are working with the integers 1 to 6, U = {1, …, 6}.
- If we are working with all integers, U = $\mathbb{Z}$.
- Both are legitimate. The problem text must say which one — without U we cannot talk about what is *outside* A.

COMMON MISTAKE / EXAM TRAP **U versus A**

- U is the whole "universe" of the problem; A is just one part (A ⊆ U). They are equal only in the special case A = U.
- Forgetting to state U is the most common reason complements go wrong (next topic).

7. Complement of a Set

DEFINITION **Complement**

Let U be the universal set and A ⊆ U. The **complement** of A (relative to U) is the set of all elements of U that are **not** in A:

$$A^c = U \setminus A = \{ x \in U : x \notin A \}$$

Read U \ A as "U with the elements of A removed".

NOTATION **Notation**

A^c and **A′** both mean "complement of A". Some books write $\bar{A}$. The backslash "\" in U \ A means "remove". The set builder says: "the x's in U that are not in A".

Key point: the complement is always taken *inside* a specified U. If U changes, so does A^c.

Complement: $A^c = U \setminus A$ (hatched region)

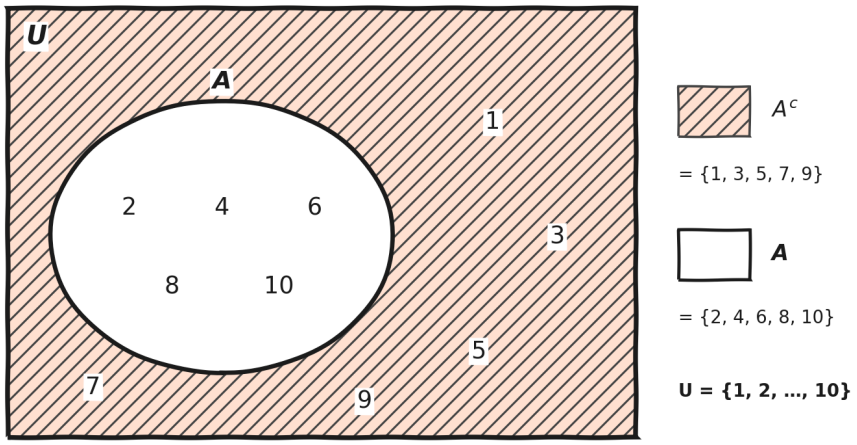


Figure 7.1 — U is the outer rectangle, A is the outlined ellipse (white). Only the hatched region — inside U but outside A — is A^c = {1, 3, 5, 7, 9}.

WORKED EXAMPLE **Example 7.1 — Finite sets**

(a) U = {1, 2, …, 10}, A = {2, 4, 6, 8, 10}.

Step 1. Go through every element of U and ask "is it in A?".

Step 2. Keep those that are not: 1, 3, 5, 7, 9. **A^c = {1, 3, 5, 7, 9}.**

Step 3. Check: $|A| + |A^c| = 5 + 5 = 10 = |U|$. In general $|A^c| = |U| - |A|$.

(b) Emails. U = {e1,…,e6} and Spam = {e5}. Then Spam^c = {e1, e2, e3, e4, e6} — the "not Spam" emails. Check: 1 + 5 = 6. ✓

(c) Dependence on U. A = {2, 4}. If U = {1,…,6}, A^c = {1, 3, 5, 6}. If U = {1,…,10}, A^c = {1, 3, 5, 6, 7, 8, 9, 10}.

WORKED EXAMPLE **Example 7.2 — Number-line complements**

(a) U = $\mathbb{R}$ and A = $(-\infty, 3)$. Everything in $\mathbb{R}$ that is not less than 3 is $x \geq 3$, so **$A^c = [3, \infty)$**.

(b) U = [0, 10] and A = [0, 4]. Elements of U above 4 are not in A, so **$A^c = (4, 10]$**.

Notice the endpoint flips: $3 \notin A$ so $3 \in A^c$; $4 \in A$ so $4 \notin A^c$. A closed end of A becomes an open end of A^c, and vice versa.

KEY IDEA **Special cases**

- $U^c = \varnothing$ (everything is in U, so nothing is left outside) and $\varnothing^c = U$.
- The complement of the complement returns the original: $(A^c)^c = A$. Check: U = {1, 2, 3, 4}, A = {1, 2}, A^c = {3, 4}, $(A^c)^c$ = {1, 2}.

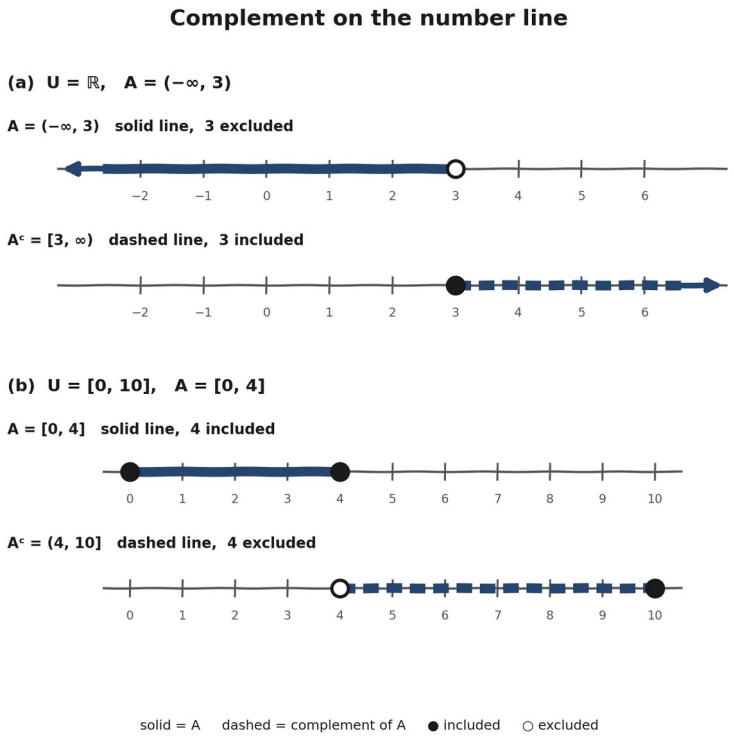


Figure 7.2 — On the number line, A (solid) and A^c (dashed) fill U with no gap between them; the shared boundary point belongs to exactly one of them.

COMMON MISTAKE / EXAM TRAP **Complement mistakes**

- **Including elements outside U.** In Example 7.1(a), $11 \notin A$, but $11 \notin U$ either, so $11 \notin A^c$.
- **Forgetting to name U.** Always say "complement relative to U = …".
- **Endpoint errors** on intervals: swap closed ↔ open at the boundary.
- **Shading wrongly** in diagrams: shade the region *outside* A but *inside* U, not outside U and not A itself.

Notation Reference Table

Symbol	Meaning	Example
{ }	braces enclosing the elements of a set	{2, 4, 6}
$\in$ $\notin$	is an element of / is not an element of	$4 \in S$, $5 \notin S$
$\varnothing$ or { }	empty set (no elements)	$\lvert\varnothing\rvert = 0$
: or |	"such that" in set-builder notation	$\{x \in \mathbb{Z} : x > 0\}$
$\mathbb{N}$ $\mathbb{Z}$ $\mathbb{R}$	natural numbers (from 1), integers, real numbers	$3 \in \mathbb{N}$, $-2 \in \mathbb{Z}$, $0.5 \in \mathbb{R}$
|S| or n(S)	cardinality (number of elements); |S| used here	$\lvert\{a, b, c\}\rvert = 3$
$\subseteq$	subset (equality allowed)	$\{1, 2\} \subseteq \{1, 2\}$
$\subset$	proper subset (this course's convention)	$\{1\} \subset \{1, 2\}$
$\not\subset$	not a subset	$\{1, 5\} \not\subset \{1, 2\}$
= ≠	equal sets / not equal sets	$\{1, 2\} = \{2, 1\}$
~	equivalent sets (same cardinality)	$\{a, b\} \sim \{1, 2\}$
U	universal set of the problem	$U = \{1, \ldots, 10\}$
A^c A' $U \setminus A$	complement of A relative to U	$\{x \in U : x \notin A\}$
[a, b] (a, b)	closed / open real interval	[0, 1], (0, 1)
∞	unbounded end (always with a round bracket)	[2, ∞)

Examination Tips and Common Mistakes

Mistake	Correct idea
Counting repeats: $\lvert\{1, 1, 2\}\rvert = 3$	Repeats are ignored: $\lvert\{1, 1, 2\}\rvert = 2$.
Writing $4 \subseteq S$ or $\{4\} \in S$	$4 \in S$ (element of set) and $\{4\} \subseteq S$ (set inside set).
Treating $\subset$ as "subset or equal"	In these notes $\subset$ is *proper*. Use $\subseteq$ when equality is allowed.
Thinking $\varnothing$ has no relation to other sets	$\varnothing \subseteq S$ for every S. But $\varnothing \in S$ is usually false.
Writing $\{\varnothing\}$ for the empty set	$\{\varnothing\}$ has one element; the empty set is $\varnothing$.
Saying equal and equivalent sets are the same	Equal $\Rightarrow$ equivalent. Equivalent $\not\Rightarrow$ equal ($\{a, b\} \sim \{1, 2\}$).
"Every infinite set is continuous"	False: $\mathbb{N}$ and $\mathbb{Z}$ are infinite but not continuous.
Mixing up $[1, 4]$ and $\{1, 2, 3, 4\}$	The interval holds *all reals* from 1 to 4; the roster holds four integers.
Computing A^c without U	$A^c = U \setminus A$ depends on U. State U first.
Putting elements outside U in A^c	Every element of A^c must belong to U.
Shading the wrong region of a complement diagram	Shade only the part of U that lies outside A.
Assuming categories are disjoint without saying so	State the modelling assumption (each email gets exactly one label).

EXAM TIP **Quick exam routine**

- Underline the **universal set** and any **inequality** (< vs ≤) in the question.
- Write sets in **roster form** before answering relationship questions; it makes checking easy.
- After finding a complement, verify $\mathbf{|A| + |A^c| = |U|}$.
- For intervals, draw a quick number line and mark **filled / hollow** dots.

End-of-Lecture Summary

- A **set** is a well-defined collection of distinct objects; order and repeats do not matter.
- $x \in S$ means "x is an element of S"; $x \notin S$ means it is not. An element and a set are different things.
- Sets can be written in **roster**, **set-builder** or **descriptive** form.
- $\varnothing$ has 0 elements; a singleton has 1; a **finite** set has a whole-number count; an **infinite** set never ends.
- **Equal** sets have the same elements; **equivalent** sets have the same number of elements.
- $A \subseteq B$: every element of A is in B. $A \subset B$ (proper): in addition $A \neq B$. $\varnothing \subseteq$ every set.
- **Disjoint** sets share no element. Email categories are disjoint if each email gets exactly one label.
- **Real intervals** are infinite sets of reals; [ ] includes an endpoint, ( ) excludes it. Not every infinite set is an interval.
- The **universal set** U is the problem's "universe"; every set in the problem is a subset of U.
- The **complement** $A^c = U \setminus A = \{x \in U : x \notin A\}$ depends on U, and $|A^c| = |U| - |A|$.

Practice Questions

Questions run from basic notation to exam-style applications. Try each before checking the answer key. Take $\mathbb{N} = \{1, 2, 3, \ldots\}$.

Part A — Notation and membership (basic)

Q1. Let S = {3, 6, 9, 12}. State whether each is true or false: (a) $6 \in S$ (b) $8 \in S$ (c) $12 \notin S$ (d) $\{3\} \in S$ (e) $\{3\} \subseteq S$.

Q2. Write in symbols: (a) "7 is an element of P" (b) "q is not an element of Q" (c) "the set A has five elements".

Q3. Which of these are well-defined sets? (a) emails labelled Spam by a filter (b) the most useful features of a dataset (c) the integers from 1 to 5 inclusive (d) big numbers.

Part B — Representing sets

Q4. Write in roster form: (a) $\{x \in \mathbb{Z} : -1 \le x \le 3\}$ (b) $\{x \in \mathbb{N} : x \text{ is odd and } x < 10\}$ (c) $\{n^2 : n \in \mathbb{N}, n \le 4\}$.

Q5. Write in set-builder form: (a) {5, 10, 15, 20, 25} (b) {2, 4, 6, …}. (c) Write "the integers greater than −3 and at most 2" in both roster and set-builder form.

Q6. A dataset D = {r1, …, r5} has ages 18, 22, 22, 35, 40 (in that order). (a) Write $Y = \{r \in D : \text{age}(r) \ge 22\}$ in roster form. (b) How many *distinct* age values are there?

Part C — Types of sets

Q7. Classify each as empty, singleton, finite with ≥ 2 elements, or infinite: (a) $\{x \in \mathbb{Z} : x^2 = 9\}$ (b) $\{x \in \mathbb{N} : x < 1\}$ (c) $\{x \in \mathbb{Z} : 2x = 8\}$ (d) $\{x \in \mathbb{Z} : x > 0 \text{ and } x \text{ is even}\}$ (e) [2, 5].

Q8. Let A = {1, 2, 3}, B = {3, 1, 2}, C = {a, b, c}, D = {1, 2, 3, 3}. Which pairs are equal? Which are equivalent but not equal?

Q9. Are the following pairs disjoint? (a) {1, 2, 3} and {4, 5} (b) {a, b} and {b, c} (c) Primary and Spam (one label per email) (d) ∅ and {1}.

Q10. Write in interval notation and say which endpoints are filled/hollow: (a) $\{x \in \mathbb{R} : 0 \le x < 1\}$ (b) $\{x \in \mathbb{R} : x > 2\}$ (c) $\{x \in \mathbb{R} : -1 < x \le 4\}$.

Q11. True or false: (a) every infinite set is an interval (b) $3.5 \in [1, 4]$ (c) $[1, 4] = \{1, 2, 3, 4\}$ (d) $(2, 2) = \emptyset$ (e) [2, 2] is a singleton.

Part D — Relationships between sets

Q12. Let A = {1, 2, 3, 4}, B = {2, 4}, C = {4, 2, 2}. True or false: (a) $B \subset A$ (b) $B \subseteq C$ (c) $B = C$ (d) $C \subset A$ (e) $A \subseteq B$ (f) $\emptyset \subseteq A$ (g) $\emptyset \in A$ (h) $|C| = 3$.

Q13. List all subsets of {x, y, z}. How many are proper subsets?

Q14. Explain, in two or three sentences, why $\emptyset \subseteq S$ is true for every set S.

Q15. For A = {1, 2, 3}, explain the difference between "$2 \in A$" and "$\{2\} \subseteq A$".

Part E — Universal set and complement

Q16. U = {1, 2, …, 12} and A = {multiples of 3 in U}. Find A^c and $|A^c|$.

Q17. U = {a, b, c, d, e} and B = {a, e}. Find B^c, $(B^c)^c$ and U^c.

Q18. A = {2, 3}. Find A^c when (i) U = {1, 2, 3, 4} and (ii) U = {0, 1, 2, 3, 4, 5}. Why are the answers different?

Q19. Find the complement: (a) $U = \mathbb{R}$, $A = (-\infty, -2]$ (b) $U = \mathbb{R}$, $A = [5, \infty)$ (c) U = [0, 1], A = (0.3, 1].

Part F — Exam-style applications

Q20. A filter gives each of 8 emails exactly one label: Primary = {e1, e5}, Social = {e2, e6, e7}, Promotions = {e3}, Spam = {e4, e8}.

(a) Write U in roster form and find |U|.

(b) State the assumption that makes the four categories disjoint.

(c) Find Spam^c and $|\text{Spam}^c|$.

(d) True or false: Promotions $\subset \text{Spam}^c$. Spam $\subseteq \text{Spam}^c$. Is Promotions a singleton?

Q21. A model outputs a probability p for "Spam". Take U = [0, 1] and the decision rule "label Spam when $p \geq 0.5$". Let S be the set of p-values that give the label Spam. (a) Write S as an interval. (b) Find S^c. (c) Are S and S^c disjoint? Is the label determined by p alone, or by p *and* the rule?

Q22. A dataset has 10 records, U = {r1, …, r10}. The set A of records labelled "positive" has $|A| = 4$. (a) Find $|A^c|$. (b) What is A^c if $A = \varnothing$? (c) What is A if $|A^c| = 0$?

Q23. Let U = {1, 2, …, 20} and $S = \{x \in U : x \text{ is prime}\}$. List S and S^c, and verify $|S| + |S^c| = |U|$.

Answer Key with Explanations

A1. (a) True — 6 is listed. (b) False — 8 is not listed. (c) False — 12 *is* listed, so $12 \in S$. (d) False — {3} is a set; the elements of S are numbers. (e) True — the only element of {3} is $3 \in S$.

A2. (a) $7 \in P$. (b) $q \notin Q$. (c) $|A| = 5$ (or $n(A) = 5$).

A3. (a) and (c) are sets: membership can be tested objectively. (b) "most useful" and (d) "big" are opinions, so they are not well-defined.

A4. (a) {−1, 0, 1, 2, 3}. (b) {1, 3, 5, 7, 9}. (c) n = 1, 2, 3, 4 gives {1, 4, 9, 16}.

A5. (a) $\{x \in \mathbb{N} : x \text{ is a multiple of 5 and } x \le 25\}$, or $\{5n : n \in \mathbb{N}, n \le 5\}$. (b) $\{x \in \mathbb{N} : x \text{ is even}\}$, or $\{2n : n \in \mathbb{N}\}$. (c) Roster {−2, −1, 0, 1, 2}; set-builder $\{x \in \mathbb{Z} : -3 < x \le 2\}$. (−3 is excluded by the strict ">", 2 is included by "at most".)

A6. (a) Y = {r2, r3, r4, r5} (ages 22, 22, 35, 40; "≥ 22" includes 22). (b) The age values are {18, 22, 35, 40}, so 4 distinct values — the repeated 22 counts once.

A7. (a) {−3, 3}: finite, 2 elements. (b) ∅: no element of $\mathbb{N}$ is less than 1. (c) {4}: singleton. (d) infinite (2, 4, 6, …). (e) infinite — an interval with $2 < 5$ contains infinitely many reals.

A8. A = B = D (same elements; repeats and order do not matter). C has 3 elements, so C ~ A, C ~ B, C ~ D, but C is **not equal** to any of them because its elements are letters.

A9. (a) Disjoint. (b) Not disjoint — b is in both. (c) Disjoint — every email has exactly one label (assumption). (d) Disjoint — ∅ has no elements, so there is nothing in common.

A10. (a) [0, 1) — filled dot at 0, hollow at 1. (b) (2, ∞) — hollow at 2, arrow right. (c) (−1, 4] — hollow at −1, filled at 4.

A11. (a) False — $\mathbb{N}$ is infinite but not an interval. (b) True — $1 \le 3.5 \le 4$. (c) False — [1, 4] contains all reals, e.g. 2.5. (d) True — no number is strictly between 2 and 2. (e) True — [2, 2] = {2}.

A12. (a) True: $2, 4 \in A$ and $1 \in A$ but $1 \notin B$. (b) True. (c) True — C = {2, 4}. (d) True — $1 \in A$, $1 \notin C$. (e) False — $1 \in A$ but $1 \notin B$. (f) True. (g) False — the elements of A are numbers. (h) False — $|C| = 2$ (repeats ignored).

A13. ∅, {x}, {y}, {z}, {x, y}, {x, z}, {y, z}, {x, y, z} — 8 subsets (2^3). All except {x, y, z} are proper, so **7** proper subsets.

A14. $S_1 \subseteq S$ fails only if some element of S_1 is missing from S. The empty set has no elements, so no such element exists and the condition cannot fail. Hence $\emptyset \subseteq S$ for every S.

A15. $2 \in A$ says the *number* 2 is an element of A. $\{2\} \subseteq A$ says the *set* {2} is a subset of A; it is true because the only element of {2}, namely 2, is in A. Note $\{2\} \in A$ is false.

A16. A = {3, 6, 9, 12}. A^c = {1, 2, 4, 5, 7, 8, 10, 11}, $|A^c| = 12 - 4 = 8$.

A17. B^c = {b, c, d}. $(B^c)^c$ = {a, e} = B. $U^c = \emptyset$.

A18. (i) {1, 4}. (ii) {0, 1, 4, 5}. The complement is taken *inside* U, so a bigger universe puts more elements outside A.

A19. (a) (−2, ∞) — $-2 \in A$ so $-2 \notin A^c$. (b) (−∞, 5). (c) [0, 0.3] — $0.3 \notin A$ so $0.3 \in A^c$; the lower end 0 is in U and not in A.

A20. (a) U = {e1, e2, …, e8}, $|U| = 8$ (2 + 3 + 1 + 2 = 8). (b) Each email receives exactly one label. (c) Spam^c = {e1, e2, e3, e5, e6, e7}, $|\text{Spam}^c| = 6 = 8 - 2$. (d) Promotions $\subset \text{Spam}^c$ is **True** ($e3 \in \text{Spam}^c$ and the sets differ). Spam $\subseteq \text{Spam}^c$ is **False** ($e4 \notin \text{Spam}^c$). Promotions = {e3} is a singleton.

A21. (a) S = [0.5, 1]. (b) S^c = [0, 0.5). (c) Disjoint — 0.5 belongs only to S. The label is fixed by p **and** the decision rule: the rule turns p into a category, and S is the set of p-values the rule sends to Spam.

A22. (a) $|A^c| = 10 - 4 = 6$. (b) $A^c = U$. (c) $|A^c| = 0$ means $A^c = \emptyset$, so A = U.

A23. S = {2, 3, 5, 7, 11, 13, 17, 19}, $|S| = 8$. S^c = {1, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20}, $|S^c| = 12$. Check: 8 + 12 = 20 = |U|. (1 is not prime, so it is in S^c.)